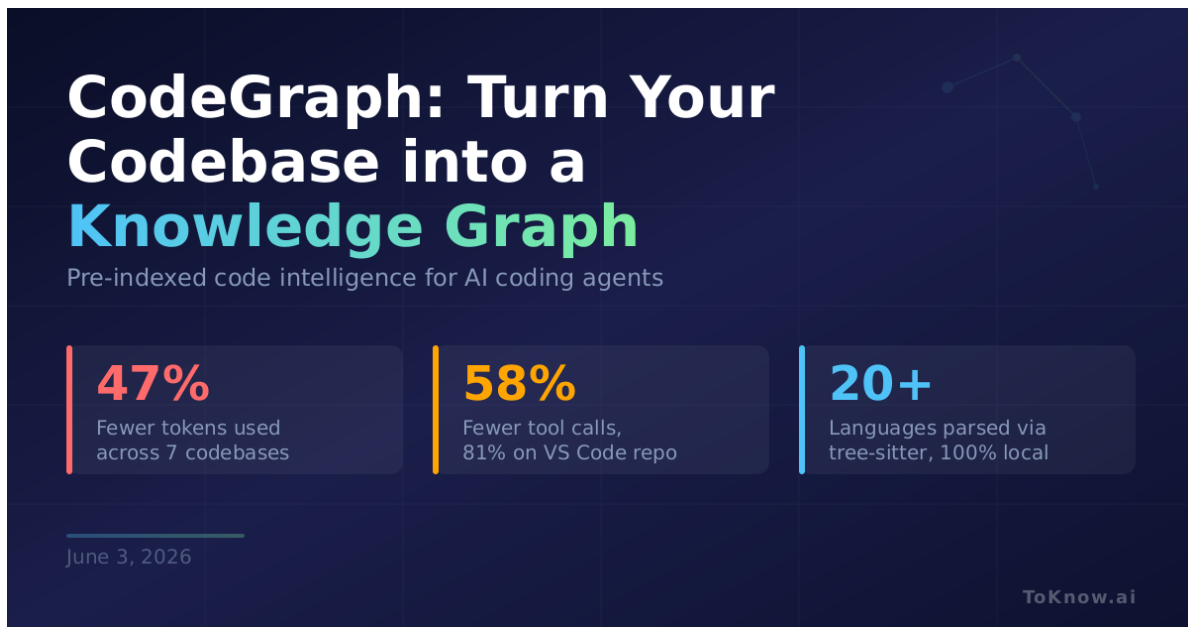


CodeGraph: Turn Your Codebase into a Knowledge Graph So AI Agents Stop Wasting Tokens

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CodeGraph: Turn Your Codebase into a Knowledge Graph
Pre-indexed code intelligence for AI coding agents

47% Fewer tokens used across 7 codebases	58% Fewer tool calls, 81% on VS Code repo	20+ Languages parsed via tree-sitter, 100% local
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ToKnow.ai

Every time you start a new AI coding session, the agent re-reads your project from scratch, burning tokens on file discovery. [CodeGraph](#) fixes this by pre-indexing your repository into a

local SQLite-backed knowledge graph that agents query through an [MCP server](#). It uses tree-sitter to parse 20+ languages, extracts symbol relationships, call graphs, and dependency edges, then serves them on demand. Benchmarked across seven open-source codebases including VS Code (~10k files) and Django (~3k files), CodeGraph cut token usage by 47%, tool calls by 58%, and wall-clock time by 22% on average. On VS Code specifically, tool calls dropped 81%. It plugs into Claude Code, Cursor, Codex, Gemini CLI, OpenCode, and Kiro with a single `codegraph install` command. A file watcher keeps the graph current as you code, and everything runs locally with no external API calls.

Tokens cost money. At Opus 4.8 pricing (\$5/\$25 per million input/output tokens), a developer exploring a large codebase burns real dollars on context loading. CodeGraph replaces those exploratory file reads with instant graph queries: a structural question like “what depends on this module?” that would require reading 20+ files gets answered in a fraction of the tokens. It also handles cross-language boundaries that trip up static analysis, bridging Swift/Objective-C calls, React Native bridges, TurboModules, and Expo modules.

AI coding agents are maturing past raw model improvements into tooling that makes each token more productive. [Dynamic workflows](#) scale agents horizontally. CodeGraph makes each individual agent session cheaper and faster by front-loading structural understanding.

Sources:

- [CodeGraph GitHub Repository](#)
- [CodeGraph Documentation](#)
- [CodeGraph npm Package](#)

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